

SME Decision-Making Pain Points in the AI Era:

A Review Analysis of Current AI Support Limits and a Scenario-Oriented Analytical Lens

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Abstract

This review-analysis article examines three linked questions: what makes decision-making difficult for small and medium-sized enterprises (SMEs), what kinds of support current AI systems can and cannot reliably provide, and whether a scenario-oriented view of decision memory offers a useful analytical lens for interpreting the gap between them. The review draws on existing literature notes in this repository covering SME decision-making, SME AI adoption and application studies, adjacent work on retrieval-augmented generation and long-term memory, and process-intelligence research on traceability and decision mining. The synthesis suggests that SME decision-making is shaped by uncertainty, short-term financial pressure, limited analytical slack, weakly structured choices, and concentrated owner-manager judgment. Current AI systems appear stronger at retrieval, summarization, bounded prediction, and procedural guidance than at reconstructing full decision situations across goals, constraints, triggering events, actions, outcomes, and evidentiary support. The article argues that fragment retrieval should not be equated with decision support in such settings. It further suggests that **scenario-oriented memory** is not yet a standard term in the literature, but may serve as a provisional analytical shorthand for combining ideas from episodic memory, workflow memory, case-based reasoning, and decision traceability. The article concludes with a narrow and defensible interpretation of the research opportunity: future work should test whether some SME decisions require more explicit episode-level context and traceability than current AI support systems typically provide.

1. Introduction

Small and medium-sized enterprises (SMEs) occupy a difficult decision environment. Compared with larger firms, they often operate with less managerial specialization, less analytical slack, lower financial buffers, and more concentrated decision authority. These constraints shape how decisions are framed, what evidence is available, how quickly alternatives must be assessed, and how much of the process can realistically be formalized. In this setting, AI support systems are often introduced with the promise of better information access, faster analysis, and more timely business insight. Yet the current literature reviewed in this repository suggests a more uneven picture.

Evidence-backed findings indicate that SME research is strongest on decision difficulty under uncertainty, working-capital pressure, uneven data conditions, and owner-manager-centered judgment (Cowden et al., 2022; Nicolas, 2021; Salles, 2006; Wong et al., 2018). By contrast, the AI-for-SME literature is stronger on adoption barriers and functional applications than on decision-grade context reconstruction (Oldemeyer et al., 2024; Schwaewe et al., 2024; Hansen and Bøgh, 2021). Adjacent literatures on retrieval-augmented generation, conversational memory, workflow memory, episodic memory, and process intelligence suggest that current systems may be able to retrieve fragments, summarize documents, or preserve limited continuity while still falling short of reconstructing a complete decision situation.

This article does not present a validated framework, a methods contribution, or an empirical system evaluation. It is a structured review-analysis essay. Its purpose is to synthesize the current evidence base

and to ask whether a scenario-oriented framing offers a useful, but still provisional, way to interpret the remaining gap between SME decision needs and current AI support capabilities.

2. Review Scope and Analytical Lens

This article is based on prior literature reconnaissance notes rather than on a new systematic search. The source base combines four types of material: SME decision-making studies, SME AI adoption reviews, adjacent AI memory and RAG studies, and process-intelligence or decision-traceability work. The source mix is deliberately asymmetric. SME-specific evidence is stronger on the business-problem side than on the memory-architecture side. The architecture side therefore relies more heavily on adjacent literatures.

Two distinctions structure the analysis. First, evidence-backed observations are separated from Working Hypotheses. Second, the article distinguishes between information assistance and decision reconstruction. This distinction matters because "useful AI" for search, summarization, or workflow aid does not automatically imply "decision-grade AI" for weakly structured business decisions.

Analytical dimension	Included in this review	Not treated as sufficient on its own	Importance
SME decision conditions	Uncertainty, cash-flow pressure, staffing limits, incomplete information, informal decision routines	Generic claims that SMEs face "complexity"	Grounds the problem in concrete managerial conditions
SME AI literature	Adoption reviews, application surveys, readiness barriers	General AI usefulness claims without decision-task specificity	Shows what the SME AI literature currently emphasizes
Adjacent AI memory literature	RAG sufficiency, conversational memory, long-term memory, workflow memory, episodic memory	Benchmark gains without a plausible bridge to business decisions	Clarifies what kinds of context mechanisms currently exist
Process-intelligence literature	Decision mining, traceability, explainable predictive operational support	Broad BPM claims without decision relevance	Provides non-LLM precedents for structured decision context

The analytical target is therefore narrower than "AI for SMEs" in general. It is closer to the question of how AI systems might support weakly structured SME decisions that depend on incomplete, cross-source, temporally distributed context.

3. SME Decision-Making Pain Points

The reviewed literature suggests that SME decision difficulty does not arise from a single bottleneck. It emerges from the interaction of data problems, operational pressures, strategic uncertainty, and organizational or cognitive limits. The recurring theme is not simply that SMEs have less data or less technology. Rather, they often have less capacity to convert imperfect information into defensible decisions under time and resource pressure.

Evidence-backed findings

- SME decisions are often made under uncertainty rather than under stable planning assumptions (Cowden et al., 2022).
- Short-term working-capital constraints can materially crowd out longer-term investment choices (Nicolas, 2021).
- Tactical and strategic decisions in SMEs are frequently weakly structured, making information requirements difficult to specify in advance (Salles, 2006).

- Smaller entrepreneurial firms often rely on less comprehensive decision routines than larger professionally managed firms (Smith et al., 1988).
- Owner goals, life events, and personal perspectives can directly shape business and financial decisions in SMEs (Wong et al., 2018).
- Review literature also points to limited skills, poor data conditions, and integration difficulty as recurring obstacles to data-driven decision making (Oldemeyer et al., 2024; Schwaeke et al., 2024; Kgakatsi et al., 2024).

Pain point category	Typical manifestation in SMEs	Why it disrupts decision quality	Evidence status in current notes	Consequence for AI support
Data problems	Partial, lagged, or weakly integrated records; irregular use of financial information; difficulty specifying information needs for weakly structured decisions	Obscures current state, hides missing variables, and makes comparison across alternatives fragile	Moderate to strong; supported by Salles (2006), Gibson (1992), Oldemeyer et al. (2024), Kgakatsi et al. (2024)	Systems must surface missing information, provenance, and uncertainty rather than assume a complete data view
Operational problems	Working-capital pressure, staffing constraints, and low organizational slack	Forces short-horizon choices, compresses analysis time, and elevates immediate constraints over abstract optimization	Strong on liquidity; moderate on staffing/slack; supported by Nicolas (2021) and Wong et al. (2018)	Support must treat liquidity, labor, and execution constraints as first-class decision variables
Strategic problems	Dynamic market conditions, weakly structured choices, and multi-objective trade-offs	Makes it difficult to stabilize goals, rank criteria, and know which information is decision-relevant	Moderate to strong; supported by Cowden et al. (2022), Salles (2006), Molina-Abril et al. (2025)	Support must help structure the problem itself, not only retrieve evidence after the problem is defined
Organizational/cognitive problems	Owner-manager centrality, informal routines, and satisficing under pressure	Concentrates judgment in a small group and leaves more of the decision state implicit or tacit	Strong; supported by Smith et al. (1988), Wong et al. (2018), Cowden et al. (2022)	Support must fit informal decision practice and make assumptions explicit without requiring enterprise-style formalization

What appears specific to SMEs, relative to larger firms, is not only scale but decision architecture. Larger firms can often distribute decision work across finance, analytics, operations, and strategy functions. SMEs more often compress those roles into a small owner-manager group.

Dimension	Typical SME condition	Typical larger-firm condition	Consequence
Decision authority	Concentrated in owners or a small leadership group	Distributed across specialized roles	Recommendations must fit owner-manager judgment
Information infrastructure	More uneven, fragmented, or manually maintained	More formalized and integrated	Support cannot assume clean enterprise data architecture
Resource slack	Lower financial and staffing buffers	Greater redundancy and analytical capacity	Decisions are more time-sensitive and constraint-dominated
Decision formalization	More informal or simplified	More formalized and process-governed	Systems should surface assumptions and gaps, not assume complete specifications

The practical implication is that SME decision support is inherently a context problem. A recommendation is only as useful as its ability to account for the constraints, history, and evidentiary limits surrounding a decision episode.

4. What Current AI Support Systems Can and Cannot Do

The literature does support a meaningful but bounded view of what AI can do for SMEs and related business contexts. In the SME-focused literature, AI is mostly associated with functional assistance: prediction, risk assessment, customer analysis, process optimization, and information retrieval. These are valuable capabilities, but they are not equivalent to reconstructing the full business situation in which a decision is made.

Evidence-backed findings

Current systems are strongest at:

- retrieving relevant documents or fragments
- summarizing material for faster inspection
- answering bounded questions over provided context
- offering predictive support in specific business functions
- maintaining limited conversational continuity
- guiding repeated tasks through workflow-like procedural support

What the current literature does not strongly demonstrate, at least in SME-specific settings, is that these systems can reconstruct the full decision episode: the operative goal, active constraints, triggering event, action path, observed outcome, and evidentiary basis for a conclusion.

Current AI support type	Typical context basis	What it supports well	What remains under-specified for decision-grade reasoning	Evidence basis
Document retrieval / RAG	Retrieved passages, files, reports, and knowledge stores	Locating relevant source material and grounding answers in accessible documents	Sufficiency of context, chronology, weighting of alternatives, and explicit rationale	Strong adjacent evidence from Joren et al. (2025) and Wu et al. (2025)
Summarization / question answering	Local document context or conversation window	Faster inspection of large volumes of text and bounded answering over supplied context	The user's operative goal, omitted constraints, and whether the available evidence is enough to decide	Moderate; supported indirectly by RAG sufficiency literature and by the absence of stronger SME-specific evidence
Functional predictive support	Task-specific business data such as sales, risk, or customer signals	Forecasting, classification, and narrow decision aids inside bounded functions	Cross-functional trade-offs, explanation of actionability, and integration with broader business context	Strong for bounded tasks in SME AI reviews; weak for full decision reconstruction
Conversational assistants with persistence	Dialogue history plus limited persistent user facts	Continuity, personalization, and repeated interaction quality	Business episode boundaries, changing constraints, and explicit links between past advice and later outcomes	Moderate adjacent evidence from long-term conversational memory work
Workflow or process-aware assistance	Reusable routines, event logs, or action traces	Repeated task execution, procedural guidance, and structured operational support	Case-specific deviations, tacit business judgment, and informal decisions outside logged workflows	Moderate to strong in adjacent workflow-memory and process-intelligence literature
Long-term user memory	Persistent preferences, facts, and user history	Stable personalization across sessions	Knowing the actor is not the same as reconstructing the decision situation facing that actor	Moderate adjacent evidence; concept clearer in agent literature than in SME DSS literature

This distinction matters because systems can appear competent in interaction while still underrepresenting the decision context that matters most. A summary of a contract, a prior conversation, or a market report

may be accurate and still not be enough for a decision that depends on timing, trade-offs, organizational constraints, and evidentiary uncertainty.

5. Why Fragment Retrieval Is Not Decision Support

The strongest direct support for this article's central concern does not come from SME-specific evaluations. It comes from adjacent work on RAG and structured decision support. The key point is not that retrieval is useless. It is that retrieval is a partial support layer, not a complete decision layer.

Sufficient Context: A New Lens on Retrieval Augmented Generation Systems is especially relevant because it shows that retrieved material can be relevant while still being insufficient for correct reasoning. In business settings, decisions often require abstention, explicit uncertainty, or comparison of alternatives rather than merely a plausible answer. **Retrieval Augmented Decision-Making** reinforces the same point by arguing that standard RAG provides context-related suggestions but remains weak on weighting, reasoning transparency, and structured support.

Dimension	Fragment retrieval	Decision-grade context
Primary unit of support	Passage, snippet, file, or isolated fact	Decision episode linking goal, constraints, trigger, action, outcome, and evidence
Typical success criterion	Relevant material is found and surfaced	A user can compare alternatives, understand trade-offs, and justify action under uncertainty
Time structure	Often weakly temporal or document-local	Explicitly chronological and sensitive to what changed over time
Treatment of constraints	Constraints may be mentioned if present in retrieved text	Constraints are modeled as active filters on what actions are feasible
Treatment of evidence gaps	Missing information can remain implicit	Missing information must be surfaced as a decision-relevant gap
User burden after system output	User still integrates fragments into a coherent situation model	System reduces synthesis burden by organizing the situation model itself

The literature reviewed so far therefore supports a narrow conclusion: fragment retrieval is helpful, but it should not be treated as equivalent to decision support when the decision depends on more than document access.

6. Related Concepts in the Literature

One difficulty in evaluating the present argument is terminological. The phrase **scenario memory** does not appear, in the current notes, to be a dominant established term. The nearest precedents are distributed across several literatures rather than gathered under one shared concept. This is both an opportunity and a risk: it suggests a cross-disciplinary gap, but it also means that the argument must avoid relabeling known ideas without adding clarity.

The closest adjacent concepts are as follows:

- **episodic memory**: memory of distinct past events or event sequences
- **workflow memory**: reusable routines extracted from prior action traces
- **case-based reasoning**: reuse of prior cases to support current decisions
- **decision traceability**: explicit record of how evidence and reasoning produced a recommendation

- **decision mining or process memory:** discovery or prediction of decisions from event logs and process traces

Concept	Primary unit of context or memory	Main analytical strength	Why it is insufficient on its own	Relation to a scenario-oriented lens
Episodic memory	Distinct past events or event sequences	Preserves temporally grounded recall and event continuity	Does not by itself specify decision criteria, evidence standards, or business constraints	Likely one component of scenario-oriented support
Workflow memory	Reusable action trajectories or routines	Captures procedure and repeatable task structure	May preserve how tasks were executed without capturing why a particular choice was justified in context	Likely useful where decisions are embedded in recurring workflows
Case-based reasoning	Prior cases linking problem, response, and outcome	Provides a strong decision-support precedent for analogical reuse	Older literature and not designed for modern multi-source LLM settings	Closest historical analogue to scenario-level reuse
Decision traceability	Evidence and reasoning path behind a recommendation	Supports auditability, explanation, and trust	Can explain a reasoning path without reconstructing the full surrounding situation	Likely a necessary property, not the whole memory model
Decision mining / process intelligence	Event-log traces and decision points in processes	Works well for structured operational decisions and explicit process states	Depends on logs and process structure that many SMEs may not possess comprehensively	Important adjacent literature for structured subsets of the problem
Scenario-oriented memory	Working Hypothesis: business decision episode as the unit of support	Integrates goals, constraints, triggers, actions, outcomes, and evidence in one analytical frame	Not an established standard term; currently a synthesis label rather than a validated category	Proposed umbrella lens for combining the partial strengths above

The closest conservative interpretation is that the literature does not yet justify a claim that **scenario memory** is an established category. It does justify treating it as a provisional analytical lens for thinking about how these existing concepts might need to be combined.

7. A Scenario-Oriented View of Decision Memory

Working Hypothesis

A scenario-oriented view of decision memory can be understood as the idea that decision support should preserve not just information fragments, but the decision episode as such. In this sense, the relevant unit is not a document, a conversation turn, or a user preference alone. It is a structured business situation that includes the operative goal, active constraints, triggering challenge, considered alternatives, chosen action, observed or expected outcome, and evidentiary basis for the reasoning.

This view should remain tentative. The current literature reviewed in this repository does not validate scenario-oriented memory as an established framework. What it does suggest is that the combination of SME decision conditions and AI support limits makes such a lens analytically useful.

Version of the thesis	Claim strength	What the current literature supports
Strong version	Decision-grade SME AI requires a combined scenario-memory architecture	Only partial indirect support from adjacent literature
Weak version	Some SME decisions require more explicit episode-level context and traceability than current systems usually provide	Plausible and more defensible given current notes
Overstated version to avoid	Scenario memory is already an established validated category for SME AI	Not supported by current evidence base

8. Discussion and Research Opportunities

The central analytical result of this review is not that current AI systems are ineffective, nor that a new memory architecture has already been justified. It is that the literature points to a mismatch: SME decisions often depend on weakly structured, cross-source, constraint-heavy context, while current AI systems are more mature on retrieval, summarization, prediction, personalization, and procedural assistance.

This mismatch suggests several research opportunities.

First, the field needs task-level evidence rather than conceptual appeal alone. It is not enough to say that richer memory would help; stronger evidence would compare fragment retrieval against richer contextual reconstruction on SME-like decision tasks. Second, the process-intelligence literature should be integrated more seriously into the AI-for-SME conversation. Event-log-based and traceability-oriented work may already solve part of the problem for structured operational decisions. Third, the literature needs better empirical work on how SMEs actually use prior episodes, partial records, and tacit knowledge when making repeat or comparable decisions.

At the same time, several cautions follow from the same review. The apparent gap may partly reflect publication lag rather than total absence of capable systems. The core bottleneck may be organizational readiness rather than memory architecture in some settings. Some terms currently in use, such as `document memory` or `long-term user memory`, may be more engineering shorthand than stable scholarly categories. And some of the strongest adjacent work still lacks direct SME validation.

9. Conclusion

The literature reviewed in this repository supports three conclusions. First, SME decision-making is difficult for reasons that are concrete and recurring: uncertainty, short-term financial pressure, limited analytical capacity, weakly structured choices, and owner-manager-centered judgment. Second, current AI systems appear better at retrieving, summarizing, predicting, and guiding bounded tasks than at reconstructing full business decision situations. Third, a scenario-oriented framing is analytically useful because it draws attention to what may be missing between fragment-level support and decision-grade reasoning.

At the same time, the review does not justify presenting scenario-oriented memory as a validated framework or mature category. The more defensible claim is narrower: for at least some SME decisions, explicit episode-level context and traceability may matter more than current support systems typically capture. If that claim holds under stronger empirical study, it would provide a more grounded basis for future design work on decision-grade AI support in SME contexts.

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